From Mindful Living to Algorithmic Dependence: Psychological Consequences of AI-Driven Health and Wellness Applications

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***Abstract*—It often goes unnoticed when switching from something that helps to something that we need. One of the latest examples of this phenomenon is connected with the influence of artificial intelligence on our daily routine. We are already starting our day with wearing a fitness tracker before our shoes and counting the number of calories eaten thanks to apps that turn food into numeric values. Previously, we would only share our secrets and thoughts with our closest people, but nowadays we can simply use chatbots for such communication. They assist in developing our competencies, including knowing about the hunger or learning to walk along an unfamiliar street and coping with emotions. These minor irritations in our daily schedule give rise to our human skills. There is a lot of evidence for a number of truths that have long been acknowledged by all of us. Among the newest terms coined due to the use of artificial intelligence technology through a sleep tracker is "orthosomnia". Apps that help keep track of calorie intake often come up in the medical history of those who suffer from eating disorders. Recent studies show that the proof of the inability of some people to recognize internal body cues, or "interoceptive awareness", prevents them from recognizing their bodily requirements because they got to know themselves through technological devices. Here are the three questions the present paper tries to answer:**
Where does the boundary lie between the stage where tracking yourself becomes counterproductive for you and starts controlling you? What is the impact on learning empathy, non-verbal intelligence, and conflict resolution from artificial intelligence companions instead of humans?
What cognitive skills will we sacrifice by that?

***Keywords*—Algorithmic Dependence, Self-Tracking Pathology, AI Emotional Support, Cognitive Offloading, Interceptive Awareness**

I. INTRODUCTION

A. The Problem

Millions of people use devices like smart watches or fitness tracking apps to determine their waking up times in the morning and when they should be sleeping and how was their sleep patterns last night. Next, they consciously decide how they should distribute their daily food intake based on the numbers shown on their smartphone application. Also, while heading to work, they use the GPS feature in their smartphones even though they have walked that same path for many years now. Lastly, instead of reaching out to their friends or acquaintances whenever they feel the heat, they turn to chatbots via an application. It is unlikely that anyone will be surprised at what happened in the above-mentioned scenario as it is just the way things are in 2025.

B. The Scale

In a short amount of time, we learned how to live in a world where we need technology to stay alive. People are using health and fitness apps quickly because they have features like tracking, alerts, tips, and suggestions. The numbers below show this. By 2025, they will also show how much we depend on AI technology.

Technology	Users	Key Finding

Wearable devices	35.8% of population	Market: $114B by 2028
AI chatbots (teens)	72% ages 13-17	For emotional support
AI relationships (adults)	28%	Romantic/intimate
Mental health AI	48.7%	Among those with conditions
Digital amnesia	91%	Use internet as brain

TABLE I: AI ADOPTION RATES (2025)

C. The Cost

It's clear that it's important to find the right balance between how much technology helps us and how much it changes how we think. We might not even know how much it costs to use these technologies. This is because users might not know about all the costs. For instance, orthosomnia is the anxiety that comes from using sleep tracking devices and affects about 2% of people [4]. We also don't know how much these things will cost us [4]. One example is that 75% of people with eating disorders use MyFitnessPal. We also don't know how much it will cost to use these devices [4]. For instance, 75% of people with eating disorders use MyFitnessPal, and 73% of those people say that the software makes their condition worse [5]. Lastly, 44% of users use their smartphones as personal memory banks [6]. A teenager committed suicide after talking to ChatGPT about harming oneself [7]. Second, an increasingly widespread trend involves cognitive anxiety. Cognitive anxiety develops due to constant monitoring by others. Nowadays, it is becoming common for individuals to rely on apps when making choices.

D. Three Necessary & Important Questions

There are three core questions that are underlined in this research paper: (1) What is the line that separates constructive health management from destructive psychological addiction in self-tracking? (2) How does artificial intelligence impact the nature of empathy and affinity in people who use emotionally supportive tools? (3) What mental abilities have been transferred to the machine? The paper will offer criteria that allow us to diagnose the problem of technological addiction, assess the effect of mechanical interaction instead of human interaction on our mind, and identify outsourced skills such as memory, attention, spatial cognition, and interoception.

II. BACKGROUND: WHAT THE RESEARCH SHOWS

A. The idea of Living Mindfully

The term mindful living can be defined as a process of living where an individual lives a life of self-awareness, decision making, and emotional stability. The techniques of mindfulness are very important for mindful living since they give the individuals a chance to evaluate their thought processes and behaviors without any bias. There is enough scientific evidence to suggest that mindful living not only helps to reduce st ress levels, but it also enhances one's concentration a bility and psychological wellness. Several studies done by scholars in recent times support that the practices of mindfulness have a connection with emotional regulation, self-awareness, and healthy behaviors. It is traditionally believed that mindfulness is intrinsically motivated. This means that the development of behaviors takes place through intrinsic motivation and discipline without any extrinsic motivation or stimuli. As per scientific evidence, self-motivation helps to sustain behavioral change and psychological stability over time. Individuals have a choice when it comes to deciding what is healthy for their wellness. .

B. Health Tracking: From Tool to Tyrant

Given the rapid advancement of technology and the existence of devices capable of collecting and analyzing data that can improve both mental and physical well-bei ng, the number of users applying different types of apps in order to help them manage health-related problems has increased significantly. Some examples of such apps include step tracking apps, calorie tracking apps, sleep tracking apps, meditation apps, as well as health professionals' recommendations offered based on the data collected from app users. According to research, us ing such applications can help app users receive more accurate information regarding their health as well as receive health care service information for those app users who do not have access to any traditional type of health care services like seeing a physician in person. Furthermore, according to research, by using wellness apps, app users can organize their day-to-day activities and track their progress towards having a healthier l ifestyle including regular physical activity and healthy eating. A sustainable commitment to living healthily can be achieved by the users through the continuous use of wellness applications and make sure that they remain compliant. In addition to tracking and reminders, most of the developers of wellness applications also collect data that is known as data-driven insights and tailor user experience. It therefore becomes easy for one to manage themselves. Wellness applications are therefore an essential part of life management today. Wearable health trackers have been popular over the last two years, and it is expected that the industry will expand its market to $114 billion in value by 2028 from $36 bil

lion recorded in 2020[1]. There are cases when some unexpected problems may result from the use of health wearables..

TABLE II: HEALTHY VS. HARMFUL TRACKING

Healthy Use	Harmful Use
Check occasionally	Check 10+ times daily
Data informs decisions	Data controls all choices
Can function without device	Panic if device unavailabl
Trust body + data	Trust only data
Improves wellbeing	Causes anxiety/guilt

Research indicates that negative effects of such tracking have been seen as well. In one case, for instance, a woman aged 27 years claimed that results from the sleep study that she did in the lab were incorrect because her Fitbit revealed poor sleep quality. This case helps us to understand the concept of orthosomnia – clinical sleep anxiety induced by sleep tracking technology, affecting about 2% of Fitbit users.[14]

C. The Emergence of AI Companions and the Loneliness Paradox

TABLE III: AI Companionship & Support Statistics (2025–2026)

AI Role	Usage Percentage
Therapist	25% - 32%
Friend	11% - 42%
Romantic Partner	16% - 33%

Around one-third of population prefers AI therapy to classical psychotherapy. Almost 24% of Americans are thinking of trying an AI therapist. 42% of college students see AI as a companion compared to only 11% of non-college young people. Concerning romantic relationships, 16% of singles had an interaction with an AI companion, with 33% among singles from Generation Z.

TABLE IV: HUMAN VS. AI EMOTIONAL SUPPORT

Humans Provide	AI Cannot Provide
Real empathy	Programmed responses
Facial expressions	Text-based only
Genuine disagreement	Always agrees
Reciprocal vulnerability	No genuine self
Unpredictable growth	Predetermined patterns
Accountability	No consequences

According to surveys conducted in six European countries, some concerns were revealed. In the study of almost 5,600 chatbot users looking for support, people felt lonely after usage, which resulted in the term “curing loneliness through isolation” [8]. Also, 9% of users reported negative responses from mental health AI tools [9]. One tragedy involved a 16-year-old boy who committed suicide after interacting with ChatGPT [7].

D. Digital Amnesia: Your Phone Remembers So You Don't

An increasingly discussed topic is the reduction of people’s memory. As a part of the Kaspersky Lab survey, it was discovered that 91% of people use the internet as a “second brain,” with 47% preferring smartphones Moreover, 44% consider smartphones as their brain substitute, and 71% do not remember their children’s phone number. Publications on digital amnesia skyrocketed recently; their quantity increased from 195 articles in 2019 to 837 papers in 2023 [24]. Neuroimaging research found that the frequent use of GPS could cause shrinkage of the hippocampus responsible for memory. Meanwhile, London taxi drivers known for their knowledge of streets have large hippocampus sizes [9]. Handwriting has proven to improve people’s ability to remember things by 34% compared to typing them.

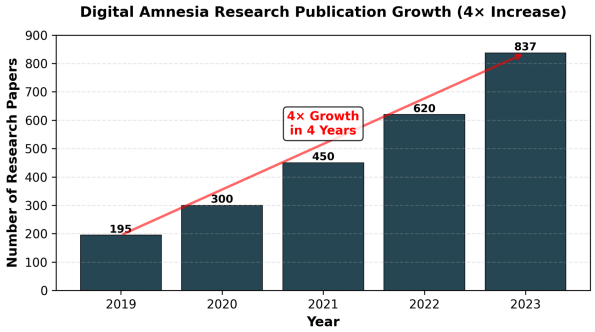


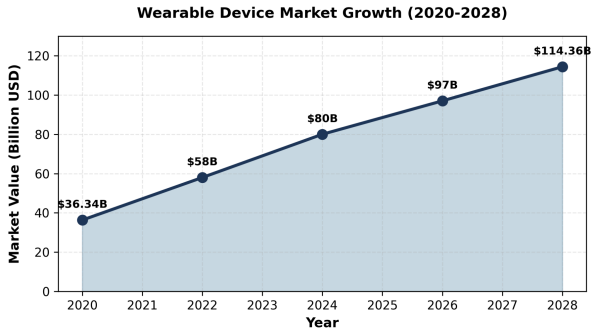
TABLE V: COGNITIVE LOSSES FROM TECHNOLOGY

Function	Loss Documented	Evidence
Phone numbers	Can't remember family	71% affected [6]
Navigation	Can't find way hom	GPS users [9]
Learning	Shallow processing	Typing vs writing [10]
Calculation	No mental math	Calculator dependency
Attention	Can't focus deeply	Smartphone presence [44]

E. Eating Disorders and Body Tracking

A strong association can be seen between body-tracking apps and eating disorders. Almost 75% of individuals diagnosed with eating disorders use MyFitnessPal, while 73% claim that it makes their problems worse. Around 71% of adults use calorie tracking apps. These applications can cause users to neglect their body's natural signs [5]. Importantly, individuals who recognize signs sent by their bodies and dismiss them are at a higher risk of having an eating disorder compared to those ignoring such sensations [29]. Overall, disbelieving one's body is more damaging than ignoring it.

III. Research Gap Identified from 15 Studies



Overall, three major topics appeared throughout the 15 studies conducted: wearables and mental health, AI chatbots, and parasocial relationships and cognitive offloading. Each topic offered many insights. However, there are gaps when considered together.

A. Findings on the impact of wearing devices on mental wellbeing vary.

Indeed, while these devices encourage physical activity, the impact on individuals' emotional state is different. On the one hand, some researchers speak about benefits such as increased consciousness and motivation. On the other hand, others describe the effect of increased stress and burnout. First, the limitation lies in the low diversity of samples that consist mostly of white adults from rich countries. Second, studies usually involve people who used devices for a short time.

Gap identified:
There is no research on the long-term and cross-cultural effects of using health-related wearable technologies on mental health.

B. AI Chatbots, Social Support, and Parasocial Relationships

When talking about AI chatbots, researchers draw an inconclusive picture. While on the one side these systems ensure emotional support and help people feel more comfortable as they give validation and company, on the other side increased reliance leads to emotional engagement and reduced communication between humans.

Multiple papers prove the importance of parasocial relationship formation in the process. These connections with artificial intelligence can become as strong as the real-life friendships, romantic relationships included. This may be beneficial when helping lonely people cope with their condition, but may turn out problematic because of potential emotional addiction.
Methodological approach:

The majority of the literature employs a relatively small sample size that includes either students or young users. While the extreme examples, such as the connection between AI companionship and severe mental disorders, are cited, they are never systematically investigated.

Gaps in the literature:
There is a lack of literature examining the psychological and social impact of the parasocial relationship with AI on long-term psychological development, specifically related to emotional dependency, identity development, and social behavior in real life.

C. Cognitive Offloading and Digital Amnesia

The results of empirical studies clearly demonstrate that there is a direct correlation between cognitive offloading and digital tools usage. While the process positively impacts the task completion, it also negatively influences memory retention. The connection was observed both in the smartphone addiction and digital amnesia studies.

However, recent works introduced a new concept of trust in technology and demonstrated that the higher levels of trust positively impacted cognitive offloading. Nevertheless, the concept is rather new, and the existing studies are limited.

Most studies focus on short-term experiments or specific groups like students. There is little evidence on how constant reliance on digital devices shapes memory, attention, and learning across different life stages.

Gap identified:
There is a lack of long-term, real-world studies examining how continuous cognitive offloading through smartphones and digital tools affects memory, learning, and cognitive independence.

D. The Overall Gap Between All Themes

When taken together, the three themes above highlight an overarching problem. Wearable devices monitor behaviors; AI chatbots impact emotions; and smartphones perform cognitive functions. Nevertheless, academic studies tend to study these digital technologies separately.

On the contrary, people interact with all of these digital technologies simultaneously. An individual may use wearable devices to track his/her sleeping pattern, communicate with AI chatbots to address emotional issues, and utilize smartphones to perform cognitive operations, all in one day.

While all of these interactions happen at the same time, there is a lack of scientific literature that investigates how these devices can interplay to affect human cognition, emotion, and behaviors.

Core research gap: There is a lack of an interdisciplinary framework to investigate the psychological, emotional, and cognitive impacts of new digital technologies.

Conclusion

Scientific studies have successfully demonstrated how specific digital technologies can affect human cognition, emotions, and behaviors. However, they do not provide a comprehensive explanation about how these technologies interplay with one another to impact humans' psychological and emotional aspects.

IV. THEORETICAL FRAMEWORK

A. Self-Determination Theory

The psychologists Deci and Ryan recognized three basic needs of humans: autonomy (the need to feel in charge), competence (the need to feel competent), and relatedness (the need to feel connected). The use of artificial intelligence negates all of the above: the algorithms take decisions on your behalf (autonomy), the individual loses the skill (competence), and the AI companion takes the place of connection (relatedness). Self-regulation theory states how individuals regulate their cognitive, emotional, and behavioral actions in order to accomplish their goals. When it comes to mindful living, the concept of self-regulation revolves around mindfulness, discipline, and personal responsibility to maintain one's physical and psychological well-being. Research indicates that self-regulation results in lower levels of stress, improved emotion regulation, and habits. Nevertheless, with the introduction of technology for wellness, certain aspects of self-regulation become externalized.

B. Enhancement vs. Replacement

There are two types of uses for tools. Enhancement: The tool amplifies your capabilities without losing your skills (using calculator when you know the math). Replacement: The tool takes over; your skills become inactive (using GPS to navigate all the time, even known routes). These two forms of technology use matter because one makes you better off and the other makes you more dependent.

C. Use It or Lose It: Brain Plasticity

Technology dependency is the psychological/behavioral

outcomes from engaging with digital devices multiple times. In health applications, consumers tend to develop habits of relying on their external assistance for decisions about their wellness routine. Your brain adapts depending on the practices you apply. Your memory center enlarged with learning routes, as with taxi drivers [9]. It became smaller due to non-practice, like using GPS [9]. When it comes to children, it may lead to permanent results if the skills do not get developed.

D. Motivation Theory

The motivation theory, especially the difference between intrinsic and extrinsic motivation, gives a better understanding of behavior change through technology. Mindful activities depend on intrinsic motivation because actions are done out of joy and personal improvement. However, well-being software may bring about extrinsic motivations, such as progress indicators, incentives, and notifications. Too much focus on external factors may make people lose their drive from inside, limit independence, and decrease compliance with good habits.

V. METHODOLOGY

A number of database searches were conducted on health tracking devices and psychology, AI chatbots and relationships, digital amnesia and memory, and eating disorders and apps. Date range: 2020-2025. Total number of articles: 15. Total number of participants: more than 80,000.
Articles should contain empirical data based on human subjects, provide answers to our research questions, be published in reputable scientific journals, and present detailed methodology and results. It is crucial to identify trends among the studies: were there several articles with similar conclusions, were samples sufficiently large, were results consistent in different cultures and age groups, and was methodology accurate?.

VI. RESULTS: ANSWERING THE THREE QUESTIONS

A. Question 1: When Does Tracking Become Harmful?

Tracking crosses from helpful to harmful when specific warning signs appear (Table V).

TABLE VI: THRESHOLDS OF HARM

Warning Sign	What It Looks Like
Checking frequency	10+ times per day
Emotional response	Panic/guilt over numbers
Body disconnect	Can't sense hunger witl app
Device dependence	Cannot function if de fails
Health impact	Tracking causes more stre

Orthosomnia affects 2% of patients [14]; 75% of people with eating disorders have tracking apps [5], while 73% report that such applications worsen their condition [5]. The process turns into a negative one if the following applies: (1) Data induces more harm than benefit; (2) you need the gadget to feel how your body works; (3) you cannot operate without a gadget; and (4) if a clinical disease develops.

B. Question 2: How Does AI Affect Our Relationships?

AI companionship eliminates experiences that build social skills

TABLE VII: EMPATHY DEVELOPMENT REQUIREMENTS

Skill Needed	Requires	AI Provides?
Reading emotion	Facial expressior tone	No - text only
Handling conflic	Real disagreemer	No - always agre
Reciprocal care	Both vulnerable	No - one-way
Nonverbal comm	Body language	No - no body

In a study conducted on 5,663 individuals in 6 nations, it was observed that chatbot users reported experiencing greater levels of loneliness rather than feeling less lonely [8]. In the context of university students, depression leads to higher usage of artificial intelligence technology, while the latter does not lead to reductions in depressive episodes [20]. The irony is that

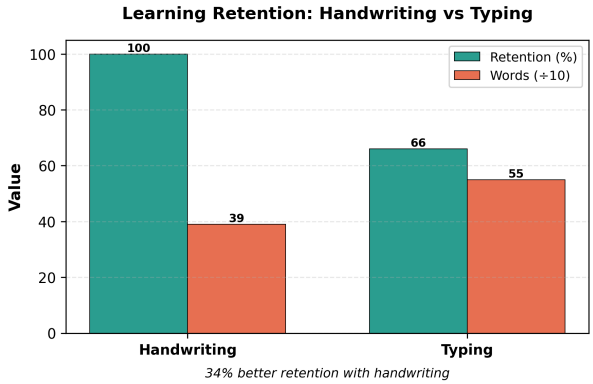
the virtual friends created by AI do make the user feel less lonely; however, they do not solve their feelings of loneliness.

C. Question 3: What Are We Losing to Algorithms?

TABLE VIII: DIGITAL AMNESIA STATISTICS

Measure	Percentage	Source
Use internet as brain	91%	Kaspersky [6]
Phone IS memory	44%	Kaspersky [6]
Can't remember kids' numbers	71%	Kaspersky [6]
Research growth (4	4x (195→837)	WoS [24]

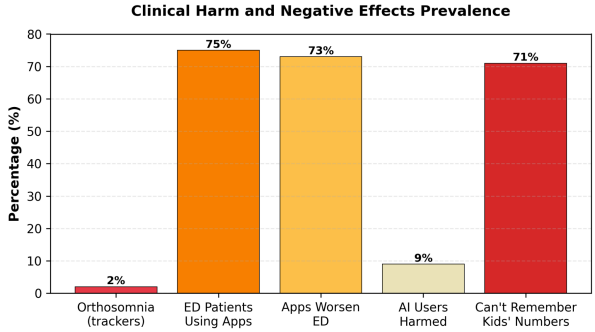
Changes in the brain documented: GPS users have shrinking of the hippocampus and poor spatial memory [9]. Taxi drivers have an enlarging hippocampus and improved memory [9]. Furthermore, handwriting leads to a 34% increase in retention compared to typing [10]. Today's children will grow up without ever developing particular skills: not navigating (born with GPS), not memorizing phone numbers (stored), not dealing with emotional distress (AI chatbot on call), not handwriting (tablet computers at school). Studies indicate that once such skills do not develop in childhood, they will never develop [12, 41].



VII. DISCUSSION

A. The Common Pattern

All three questions reveal the same underlying problem: short-term gain leads to long-term loss. Tracking feels helpful but causes anxiety and disconnection. AI feels supportive but prevents real relationship skills. Devices feel convenient but abilities atrophy.



B. Who's Most at Risk

TABLE IX: VULNERABLE POPULATIONS

Group	Why Vulnerable
Teenagers	Brain developing; 72% use AI
Mental health conditions	48.7% use AI; 9% harmed
Eating disorder history	75% use tracking; 73% worse
Low income	Can't afford human services

C. What Can Be Done

For the individual: (1) Put yourself to test - can you manage without your device for 24 hours straight? If no, then the dependency phase has been reached beyond the use phase. (2) Exercise your abilities; use your brain - don't rely on GPS, memorize phone numbers, take notes on paper, reflect emotionally. (3) Set boundaries - reduce tracking check-in frequencies; focus on human interaction.

For designers: reduce frequency of checking, avoid addiction triggers, such as streaks; sometimes challenge instead of agree; disclose limitations; warn of overuse.

For governments: consumer protection measures for health applications, age limitation on AI companionship, mandatory crisis response measures, digital literacy education programs, funding human resources such as counselors and health care providers.

VIII. CONCLUSION

Technology should be used to support mankind, not to replace it. However, we are gradually getting there.

In this paper, the literature review regarding the use of health and wellness applications was conducted with

respect to the psychological effects of these technologies. The findings have revealed that while wellness applications help improve the level of adherence to activities and mindfulness practices, as well as provide greater opportunities for accessing psychological services, they may affect the process of self-motivation, self-regulation, and independent behavior. The person may face stressful situations when applying wellness technologies in a goal-oriented manner. According to the research, it can be concluded that the implications of wellness applications are dual in nature. On the one hand, wellness applications can be effective means to promote healthy behavior. On the other hand, they may undermine mindful existence when overused. Therefore, future applications should support technology-mediated guidance and self-motivation.

You should think over and reflect on the following questions before letting artificial intelligence make the next choice for you: (1) Is it impossible for me to do it myself? (2) Or am I just avoiding something that I find unpleasant? (3) What do I not learn? (4) What am I going to miss if I do not learn it? It is crucial to understand the significance of losing something being unable to do things in an optimal way, losing one's humanity, and being subject to the process. On the other hand, we can benefit from all that is accessible to us, keep our humanity, and remain in charge of the process. The most serious risk is connected to children who may have irreparable harm inflicted on them since they cannot use their ability to navigate, remember, empathize, and regulate emotions. Scientific studies confirm such assumptions [12, 41]. Is it necessary to use artificial intelligence? But there is another question to consider before making any decisions: Are we capable of trusting our instincts to tell us whether we should use artificial intelligence or not?

Restrictions consist of those that have been ignored: (1) Long-term effects – most information is short-term in nature. What can happen in a case where an individual develops in his childhood years under the influence of AI? (2) Causation – does AI cause mental conditions or do people who already suffer from mental problems prefer AI? In all likelihood, both are true. (3) Cultural differences – research has mainly been carried out in Westernized societies, which does not mean much for other countries. (4) Individual variations – why are certain individuals more prone to developing mental problems than others?

Essential: long-term experiments during 20 years, brain scanning that would suggest recovery times, interventions testing experimentations, global studies, and improved diagnostics. Moreover, users may evaluate their dependency on AI through a specially-designed questionnaire for this research paper. It is accessible at the following link:
https://assesment-researchpaper.netlify.app

IX. Future Research

In regards to future research directions, it should be noted that it is important to take into consideration all of the limitations of the current study. In particular, it is critical to emphasize the necessity for longitudinal studies aimed at evaluating the effect of long-term exposure to technologies on developing self-regulatory abilities, intrinsic motivation, and addiction. Individual differences including personality traits, stress level, and ability to engage in mindfulness should not be overlooked. It will be very interesting to examine how the effectiveness of apps aimed at wellness such as physical activity apps, meditation apps, and nutrition apps differ from one another in enhancing psychological well-being. The combination of both qualitative and quantitative data in the form of mixed methods is highly encouraged. Finally, it is essential to focus on developing solutions aimed at encouraging more mindful use of technologies.

X. Acknowledgement

We would like to convey our sincere thanks to Chitkara University, Patiala, for giving us the opportunity to study under its tutelage. We owe our sincere thanks to our mentor, Dr. Rajat Takkar, for guiding us in the process of developing this research paper. The inputs that he provided have played a significant role in making this paper what it is today. We also acknowledge the researchers and scholars who have provided the foundation for this research through their writings and research papers on which we have developed this research paper. Their contributions have helped us comprehend the psychological and emotional impact that the implementation of AI in the health and wellness sector can create.

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